Cocoyam Aggregation and Phased Farm Business Plan

Prepared for: Grainy Palace Farm Limited

Planning basis: Ghana; 20-acre cultivation ambition; off-taker agreement reported by the owner

Status: Decision-ready framework. Confirm the commercial terms and local costs before committing funds.

Recommendation

Build a **buyer-backed cocoyam aggregation business first**, and add own cultivation in measured blocks. Do not spend scarce capital planting all 20 acres at once. Use the off-take relationship to generate early trading cash flow, learn the buyer's grading and delivery requirements, and establish farmer supply relationships. Negotiate the 248-acre land opportunity as a documented, phased lease that allows the company to begin with a 2–5 acre pilot and expand toward 20 acres only after the first crop cycle and cash-flow gates are met.

This is a combined **aggregation plus nucleus-farm model**: smallholders provide near-term volume; Grainy Palace Farms grows a controlled share of supply and develops production knowledge. It fits a capital-constrained start better than a farm-only plan because cocoyam cultivation requires cash before harvest, while aggregation can turn over working capital sooner if the buyer pays promptly. The purchase agreement is an advantage only to the extent that it defines volume, quality, price, delivery and payment clearly.

Business concept

Purchase marketable cocoyam from contracted smallholders, grade and consolidate it, and deliver to the off-taker against written purchase orders or confirmed delivery schedules. In parallel, lease and plant a small initial area using the same buyer specification. Keep the two activities in separate ledgers and cost centres so the company can see whether aggregation and farming each make money.

Customer: the contracted off-taker.

Suppliers: smallholder farmers, with the company’s own pilot plot as a later additional source.

Value offered: reliable consolidated supply, consistent grading, traceable farmer lots, and planned deliveries.

Do not assume a guaranteed market merely because an agreement exists. Confirm whether it is binding or non-binding, whether there is a minimum quantity, how price is set, what happens when quality is rejected, and how quickly accepted deliveries are paid.

Why phase the farm

Ghana-focused research places typical cocoyam corm yields around **6.2–6.5 tonnes per hectare** in surveyed production settings. At that indicative range, 20 acres (about 8.09 hectares) would imply roughly **50–53 tonnes of corms** in a mature harvest, before grading losses. This is a planning reference, not a forecast: yields depend on variety, planting material, soil, rainfall, management and harvest timing, and research results should not be treated as a guarantee for this parcel. Labour is a major production cost in Ghana cocoyam studies. The company should therefore validate the site and actual labour/input budget with the district MoFA extension officer before scaling. KNUST Ghana production study, CSIR Ghana production survey.

At the same time, the crop is vulnerable to post-harvest deterioration. FAO guidance says ambient-stored cocoyam can keep only about 4–6 weeks without serious losses under warm conditions. Start with quick collection-to-delivery cycles and ventilated, shaded handling; do not finance speculative stockpiling until losses and buyer demand are measured. FAO storage guidance.

Commercial model and operating cycle

1. Obtain the off-taker's written rolling demand forecast, order schedule, specification and current buying price.

2. Register and map supplying farmers; record expected harvest dates, estimated volumes and agreed quality standards.

3. Collect only against a confirmed buyer order or a tightly controlled delivery plan. Where practical, inspect and weigh at source.

4. Grade, remove damaged or undersized product according to the buyer's written standard, and record farmer, weight, grade, date and purchase price for every lot.

5. Consolidate transport and deliver promptly. Obtain signed weight/acceptance records and reconcile rejected quantities immediately.

6. Pay farmers according to transparent terms. Where cash is short, negotiate buyer deposits, direct payment to farmer groups, or short-cycle purchase-order finance before promising farmer payment dates.

7. Record actual contribution margin and cash days for each delivery; use proven margins to fund the next procurement cycle and the farm pilot.

Keep procurement money separate from lease and farm-development money. Never use money committed to pay farmers for land clearing or planting.

Phased implementation

Phase	Timing / gate	Actions	Capital rule
0. Verify and secure	Before lease or large purchases	Review the off-take contract; obtain buyer forecast and specification; verify the landowner's title/authority, boundaries, access, water/drainage, and lease terms; request local production and transport quotes.	Spend only on due diligence and costs needed to validate an initial trading cycle. Do not pay for or commit to all 248 acres.
1. Prove aggregation	First 1–3 delivery cycles	Start with a manageable farmer network and confirmed orders; trial grading, weighing, transport and payment reconciliation. Keep batch records and measure rejection, losses, gross margin and days-to-cash.	Prefer buyer pre-finance/deposit, buyer-paid transport, direct farmer-group payment, or supplier credit tied to a confirmed order. Limit purchases to the cash cycle the business can fund.
2. Pilot own production	First planting season after site check	Lease/occupy 2–5 acres initially (or the smallest block with viable site economics); use quality planting material; plan labour, weed control, drainage and harvest against local extension advice and buyer specs.	Fund the complete pilot budget, including contingency and harvest/transport, before planting. Do not count an unharvested crop as working capital.
3. Expand to 20 acres	After pilot harvest and cash review	Expand in blocks toward 20 acres if the pilot meets yield, quality, cost and buyer acceptance targets and the aggregator continues to turn cash profitably.	Expand only with cash retained from operations, suitable seasonal credit, or committed project finance; maintain a separate operating reserve.

If the landowner insists on immediate rent for all 20 acres, negotiate a short initial term, staged rent/payment, or a written option on the balance. If the only offer requires a large non-refundable upfront payment for the full area, defer cultivation and continue aggregation until capital is available.

Working-capital and funding plan

The immediate financing need is principally **short-cycle procurement working capital**, followed by a separately budgeted farm establishment cost. Seek finance in this order:

1. Off-taker support: deposit or advance against confirmed purchase orders; prompt payment terms; buyer-provided bags, collection, or transport; or payment directly to farmer groups after acceptance.

2. Farmer-group arrangements: pre-agreed harvest quantities and transparent pricing; where inputs are advanced, use written farmer agreements, capped advances, and repayment at delivery. Avoid tying up funds in open-ended input credit.

3. Supplier and logistics terms: negotiate payment after delivery/acceptance for transport, bags and handling, matched to the buyer's settlement date.

4. Formal value-chain finance: prepare a contract, delivery history, farmer register, cash-flow schedule and records for a bank or appropriate agricultural finance programme. Ghana's OVCF describes support for outgrower schemes and value-chain integration; check current eligibility, participating lenders, terms and availability directly before including it in the funding base case. OVCF programme information.

5. Owner/equity capital or grant: reserve for due diligence, pilot establishment and business systems, not speculative inventory.

Avoid expensive unsecured borrowing to fund the entire farm before verifying the land and market. Do not sign personal guarantees or borrow against family assets without reviewing repayment timing against the crop cycle and buyer's actual payment behaviour.

Financial plan: fill with verified local quotations

No defensible total startup budget or profit forecast can be calculated from the information currently supplied: the purchase price, sale price, demand volumes, delivery frequency, rent, land preparation, labour, seed corm cost, inputs, transport, rejection rate and payment delay are unknown. Obtain written local quotes and complete the following before asking a lender or investor for a fixed amount.

Aggregation unit economics (per delivery lot)

Item	Calculation
Accepted tonnes	Tonnes purchased × (1 – measured rejection/loss rate)
Sales revenue	Accepted tonnes × buyer price per tonne
Product cost	Tonnes purchased × farmer purchase price per tonne
Variable delivery cost	Transport + loading + bags/packaging + weighing/market charges + handling
Contribution before overhead	Sales revenue – product cost – variable delivery cost – finance cost
Working capital required	Farmer payment + trip/handling costs + any advances until buyer cash is received
Cash conversion days	Days from paying farmers to collecting cleared buyer funds

Proceed with a lot only when the forecast contribution is positive after realistic rejection, transport and finance costs, and the company can honour farmer-payment terms through the full buyer payment delay.

Farm budget (per acre and for the chosen pilot block)

Quote separately: lease and legal/title checks; site clearing; ridging or other land preparation; planting material and delivery; planting labour; weeding and crop care; pest/disease management; water/drainage work if required; field supervision; harvesting; grading; bags/crates; transport; and contingency. Record which costs are one-time and which repeat each season. Do not extrapolate a 20-acre cost from an old published Ghana margin: input and wage prices have changed, and this parcel's conditions are unknown.

Funding request formula

Requested facility = peak net procurement cash gap for confirmed orders + fully funded pilot farm budget + agreed operating reserve – buyer advances/owner contribution.

Size procurement finance from the maximum outstanding cash gap across a delivery cycle, not from annual sales. Keep the farm pilot budget ring-fenced so harvest and delivery can still be paid for if aggregation cash is temporarily tied up.

Key risks and controls

Risk	Control
Off-taker demand or price changes	Written orders/forecast, price formula and acceptance rules; cap procurement to confirmed demand.
Buyer delays payment	Agree payment days and late/dispute process; obtain deposit or direct farmer payment; limit exposure to one unpaid delivery.
Quality rejection and disputes	Buyer-approved written grading specification; sample inspection; calibrated weighing; signed acceptance at delivery.
Post-harvest loss	Harvest/collect to order; ventilated shade; fast transport; daily stock aging and loss records.
Farmer side-selling or short supply	Build relationships with multiple farmers; transparent pricing and prompt payment; realistic, written commitments.
Land/title or water suitability problems	Independent document and boundary checks; visit during/after rains; confirm access, tenure, drainage and water before lease payment.
Crop yield below plan	Pilot block, extension support, sound planting material, field records; expand based on measured harvest only.
Working-capital shortfall	Separate accounts; cash-flow forecast by delivery; reserve; no farm spending from farmer-payable funds.
Input and price inflation	Get current written quotes; refresh budget before each season and purchase cycle.

Management dashboard

Review weekly during procurement and at each farm milestone:

- Tonnes ordered, purchased, accepted and rejected; on-time delivery rate.
- Purchase cost, delivered cost and contribution per tonne / per delivery.
- Farmer payment days and buyer cash collection days; outstanding receivables and payables.
- Post-harvest loss percentage and reasons for rejection.
- Farm acres planted, labour/input cost per acre, crop establishment and expected harvest date.
- Pilot tonnes harvested per acre, marketable share, delivered cost per tonne and buyer acceptance.
- Cash available for the next cycle after reserving farmer payments, taxes, debt service and farm completion costs.

Immediate 30-day action list

1. Obtain and review the signed off-take agreement and ask the buyer for a written 3–6 month volume forecast, grade/size standard, price basis, rejection procedure, delivery point, transport responsibility and payment timeline.

2. Request three current local quotes each for land preparation, planting material, labour, transport and lease terms. Ask the district MoFA extension office to inspect the proposed site and advise on variety, planting calendar, soil/drainage and a realistic yield range.

3. Verify the landowner's authority to lease; commission boundary and title/land-record checks; negotiate phased possession and payment with clear renewal/termination, access, water, improvements and dispute terms.

4. Identify and register an initial group of farmers sufficient for the first confirmed order; document farm location, expected harvest, volume and quality agreement.

5. Run one small delivery cycle and reconcile weights, rejection, margin and payment days before increasing volume.

6. Build a 12-month monthly cash-flow forecast with base, low-yield/low-price and delayed-payment cases. Do not submit a fixed financing request until it reconciles to the contracts and quotations.

Information needed to turn this framework into a lender-ready forecast

- Copy/key terms of the off-take agreement: buyer, minimum/maximum tonnes, price, grade, delivery schedule, acceptance, payment days and remedies.
- Expected demand per week/month and delivery location/transport responsibility.
- Farmer purchase prices and locations; current market price checks.
- Lease price per acre, deposit, duration, payment schedule and exact parcel location.
- Site soil, slope, drainage, water/access, land history and planting season.
- Pilot area decision (2, 3, or 5 acres), variety, planting-material source, and extension officer's budget/yield estimate.
- Current costs for inputs, labour, aggregation, packaging, transport and insurance, if available.
- Owner contribution, available equipment, existing liabilities and proposed finance terms.

Sources and planning limits

This plan is a commercial framework, not a promise of returns. Ghana yield references above come from older or survey-based research and are useful only for initial scenario setting. The plan deliberately does not present an invented Ghana farm-gate price, rent, startup capital requirement, or profit forecast. Replace assumptions with signed buyer terms and current quotations for the actual district and season. Have a Ghanaian accountant and lawyer review the final cash-flow model, lease and farmer/off-taker contracts before taking debt or signing a long lease.